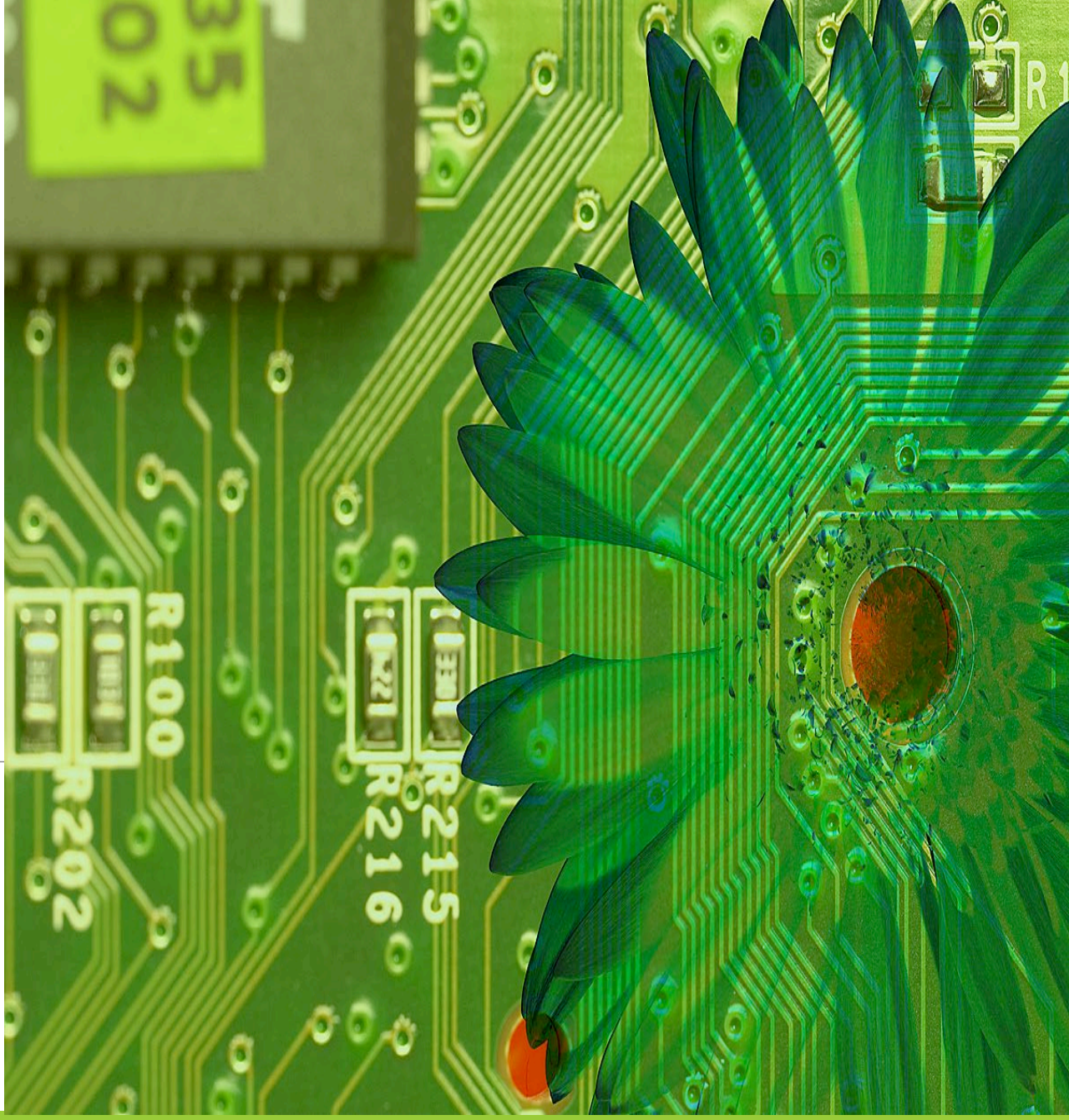
Green ITSM



JÖNKÖPING UNIVERSITY
School of Engineering

Background

Sustainable development is the kind of development that meets the needs of the present without compromising the ability of future generations to meet their own needs (Brundtland, 1987)

The Fit

- The **Social** dimension
- The **Ecologic** dimension
- The **Economic** dimension

The Fit

- The **social** dimension could be viewed as:
  - "The objective"
- The **ecologic** dimension could be viewed as
  - "The limits"
- The **economic** dimension could be viewed as
  - "The means"

THE GLOBAL GOALS
For Sustainable Development

blueprint to achieve a better and more sustainable future for all

https://www.un.org/sustainabledevelopment/sustainable-development-goals/

About the Problems

- One of the **Grand Challenges** (e.g., UN 2019)
- The three dimensions (social, ecological and economic) interact but they simultaneously **limit** each other.
- The problem is not simple it **requires comprehensive solutions**
- The problem is very complex! The problem **requires cooperation** between different actors in society.
- The problem is **global**
- "We sleep **well** at night“ (We should not if we care…) (c.f. Gray 2012)

“Organisations are **urged** to consider their **carbon footprint** and redesign business processes to reduce carbon emissions (Ghose et al. 2009).”

Carbon footprint = total set of greenhouse gas emissions caused by an individual, event, organisation, or product,

How does ITSM relate to Sustainability?

Green IT(SM)

- According to Murugesan (2008), Green IT refers to the study and practice of designing, manufacturing, using, and disposing of computers, servers, and associated subsystems efficiently and effectively with minimal or no impact on the environment. He states the objective of Green IT is "**to achieve economic viability and improved system performance and use, while abiding by our social and ethical responsibilities**" (p. 26).
- Many of the Green IT issues and activities identified by Watson et al. (2007) and Murugesan (2008) fall squarely in the domain of ITSM, in particular:
  - purchase of energy-efficient computing equipment;
  - power management to reduce energy consumption of data centres and replacement of personal computers with energy efficient thin clients;
  - consolidation and use of virtualisation software to run multiple operating systems on one server; and
  - recycling and responsible waste disposal of obsolete computing equipment.

Impact of IT on the Environment

IT has been contributing to environmental problems in ways that that most people do not realize.

Computers and other IT infrastructure consume significant amounts of electricity, which is increasing day by day, placing a heavy burden on our electric grids and contributing to greenhouse gas (GHG) emissions.

Moreover, IT hardware poses environmental problems during both its production and its disposal.

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Environmental Impacts of IT

- The increased number of digital devises and their use, along with their frequent replacements, make IT's environmental impact a major concern.

- Total electrical energy consumption by servers, computers, monitors, data communications equipment and data centre cooling systems is steadily increasing.

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Environmental Impacts of IT

- "ICT industry has a significant impact on the environment with estimates that it contributes carbon dioxide emissions equivalent to the aviation industry (ACS 2007; Gartner 2007). "
- ICT has a significantly negative effect on carbon dioxide emission" (Lu 2018)
- It is recognised that the global ICT industry has a significant impact on the environment with estimates that it contributes carbon dioxide emissions equivalent to the aviation industry (ACS 2007; Gartner 2007).

  Large organisations spend between four and ten percent of their total ICT budgets on energy and this proportion is expected to quadruple by 2012 due to the twin factors of power-hungry hardware and rising energy costs (Gershon 2008).
- Today, Green IT seems to be one area where practice is leading academia and researchers need to study the innovations put in place by practitioners.

Or…digital technology as a Saviour For a Better Planet

Whilst many people consider IT to be part of the problem to environmental pollution, it can be its saviour too.

IT is both a **solution and** a **problem** for environmental sustainability. he promotion of ICT becomes one of the important strategies introduced to mitigate carbon dioxide emissions for various countries (Lu 2018)

……

Additional Resource: Green IT: a saviour for a better planet, San Murugesan, June 2011, http://sation.in/2011/06/green-it-san-murugesan-interview/

How IT Can Help Create a Sustainable Environment?

OPERATIONAL LEVEL

How IT Can Help Create a Sustainable Environment?

IT can help create a more sustainable environment by:

1. Coordinating, reengineering and optimizing the supply chain, manufacturing activities and organizational workflows to minimize their environmental impact.
2. Making business operations, buildings and other systems energy more efficient.
3. Helping decision making by analysing, modelling and simulating environmental impacts.
4. Providing platforms for eco-management and emissions trading;
5. Auditing and reporting energy consumption and savings; and
6. Offering environmental knowledge management systems and decision support systems.

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Green IT 1.0 and 2.0

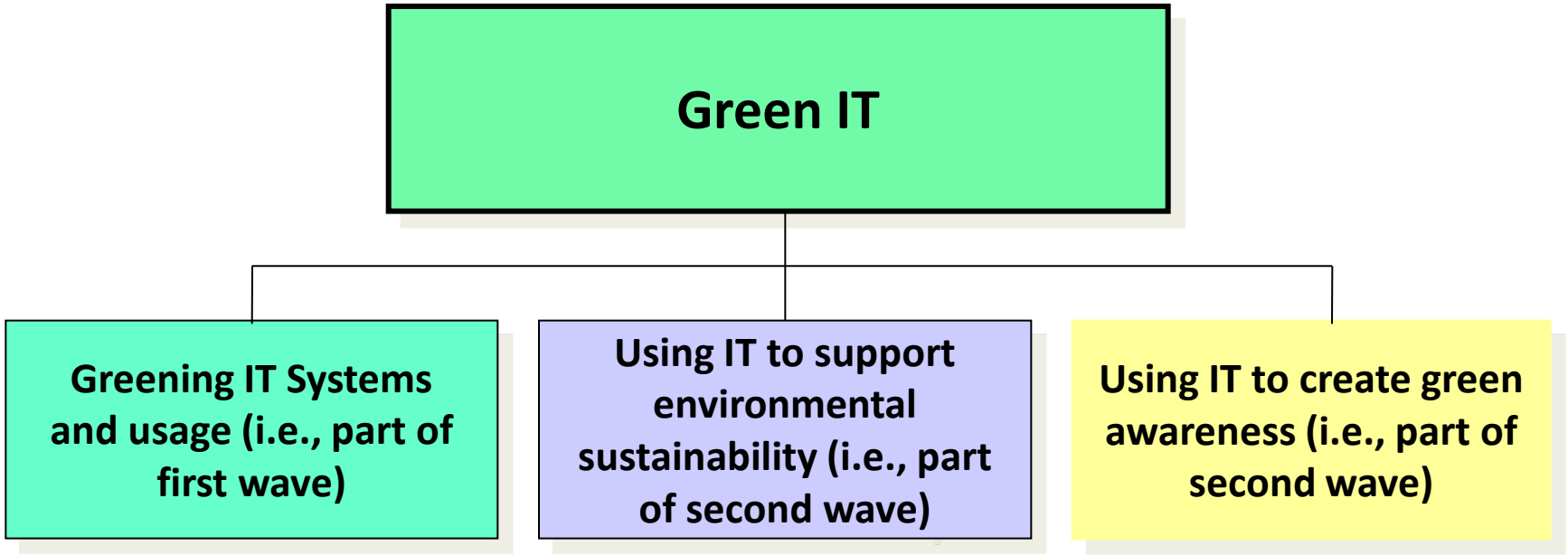
Greening of IT, or Green IT 1.0 (The first wave of green IT)

It's internally focussed on reengineering IT products and processes to improve IT's energy efficiency, maximize its use and meet compliance requirements.

Greening by IT, or Green IT 2.0 (The second wave of green IT)

It's externally focussed and empowers a range of other green initiatives aimed at reducing environmental degradation and reducing GHG emissions. For example, the second wave is focused on environmentally sound business transformation, IT-based sustainability innovation, sustainability-based IT innovations and enterprise-wide sustainability.

Green IT Dimensions



Green ITSM

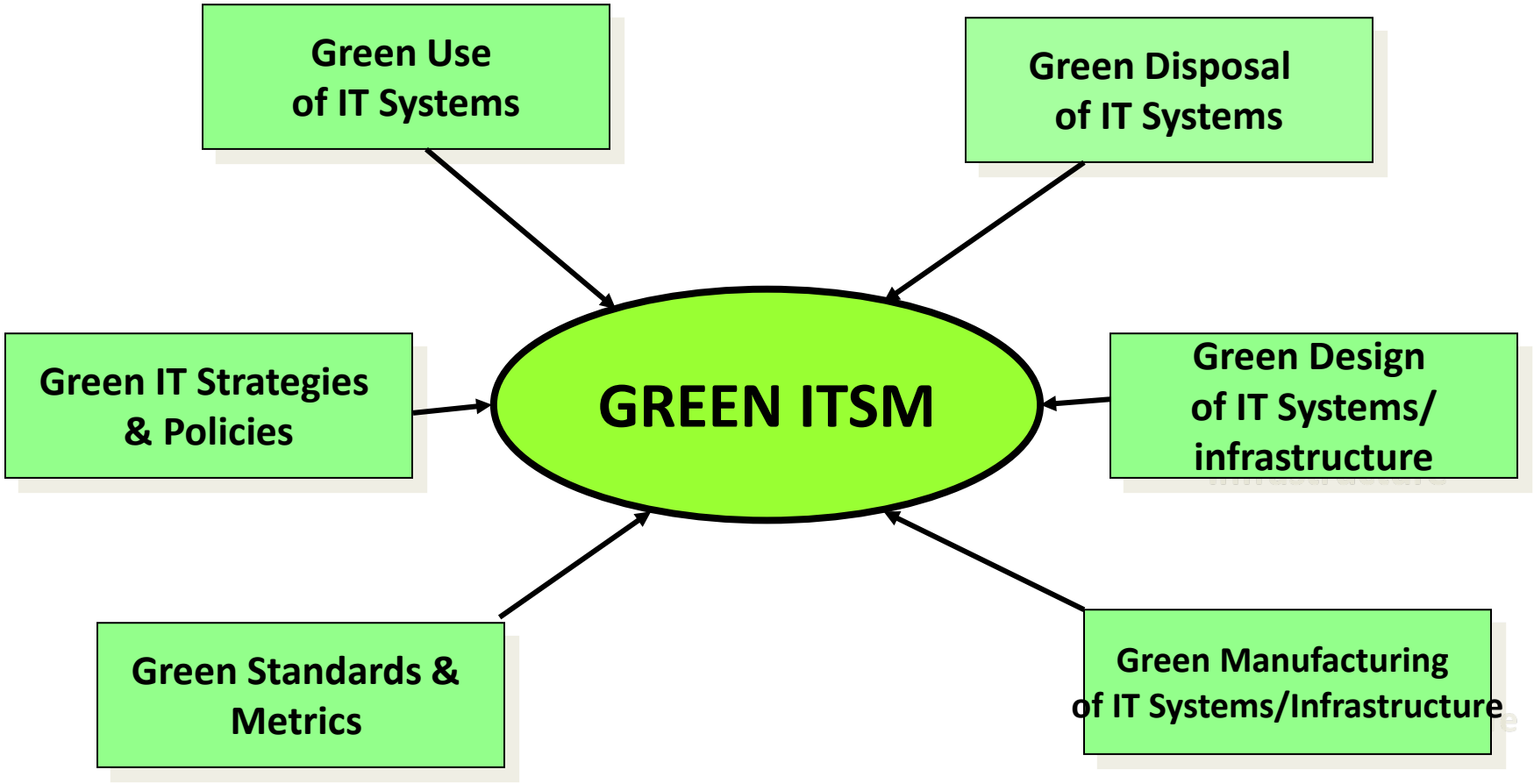
"ITSM has an important role to play in greening IT as it **controls ICT infrastructure** which consumes vast quantities of energy in manufacture of components as well as in the ongoing operation of ICT systems (Erek et al. 2009; Firth 2008). "

"IT Operations and Services form a vital subset of the ICT industry and **can enable sustainable service**. Personnel who provide IT Service Management (ITSM) include in-house ICT departments delivering ICT services within organizations as well as vendor service providers of out-sourced services. "

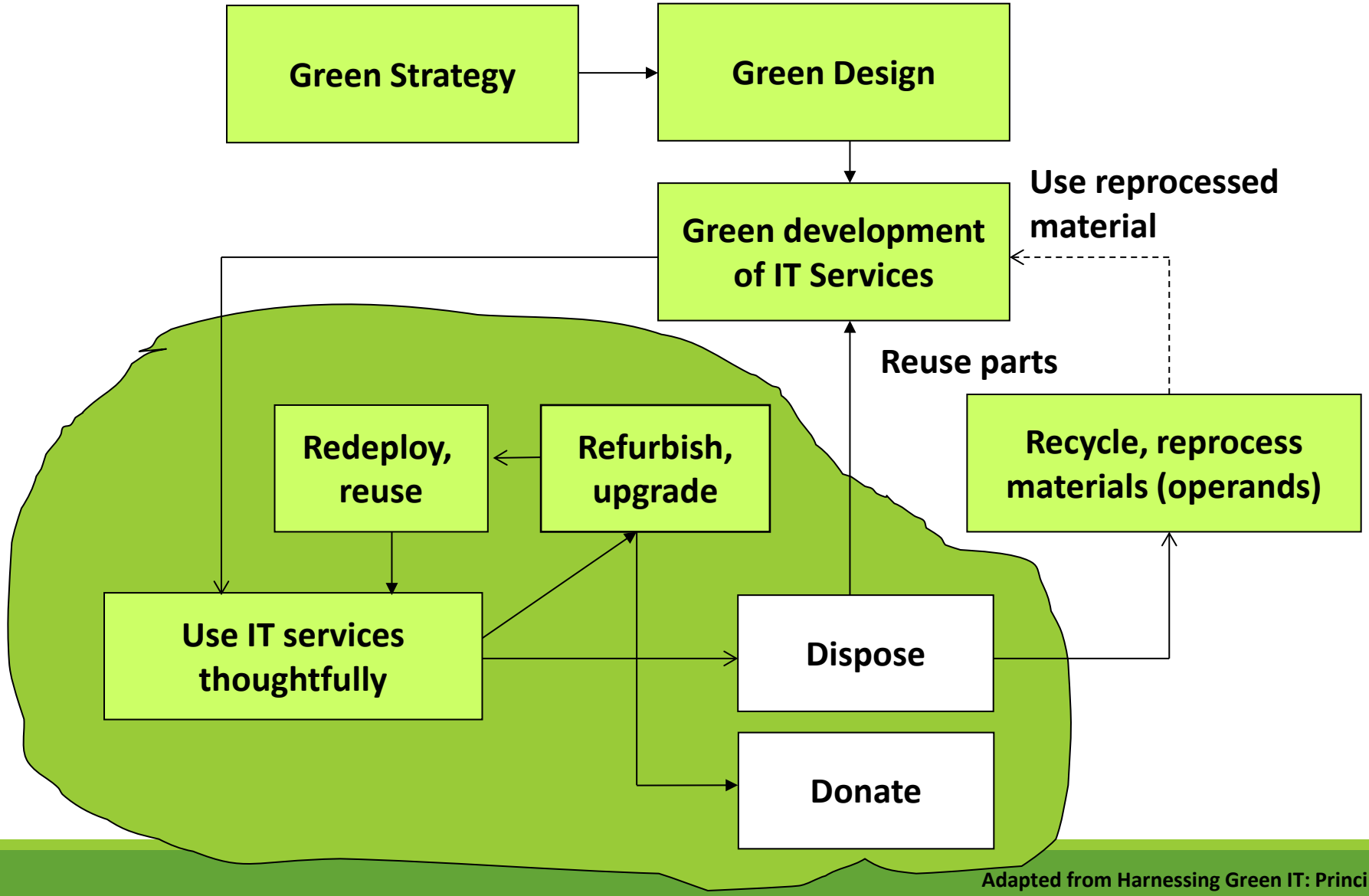
ITSM is also responsible for the **disposal of upgraded ICT equipment** which contributes significant quantities of toxic waste to the environment (Blanke 2009).

ITSM can manage **the whole green service lifecycle** (murugesan, 2012)

Model of Green ITSM



Greening the ITSM's Lifecycle



Adapted from Harnessing Green IT: Principles and Practices
San Murugesan & G.R. Gangadharan (Editors) © 2012 John Wiley & Sons

How ITSM Can Become Green

Moreover, ITSM can support organizations to design and manage services to improve:

1. **green use:** reducing the energy consumption of computers and other information systems as well as using them in an environmentally sound manner
2. **green disposal:** refurbishing and reusing old computers and properly recycling unwanted computers and other electronic equipment
3. **green design:** designing energy-efficient and environmentally sound components, computers, servers, cooling equipment, and data centers
4. **green manufacturing --** manufacturing electronic components, computers, and other associated subsystems with minimal impact on the environment

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E-Waste

- **E-waste**: Discarded computers and electronic **goods**.
- It is a fastest-growing waste type and poses environmental problems.
  - The United Nations Environment Program estimates that 20–50 million tons of e-waste are generated worldwide each year, and this is increasing.
  - IT hardware contains toxic materials like lead, chromium, cadmium and mercury.
    - If e-waste is not discarded properly, it can harm the environment and us. I.e., we bury IT hardware in landfills, toxic materials can leach harmful chemicals into waterways and the environment. If burned, they release toxic gases into the air we breathe.

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The Three Rs of Green IT (operand focus)

Unwanted computers, monitors and other electronic hardware should not be thrown away as rubbish, as they will then end up in landfills and cause serious environmental problems.

Instead, we should refurbish and reuse them, or dispose them in environmentally sound ways;

Reuse, Refurbish and ***Recycle =*** the three 'Rs' of greening unwanted hardware.

Mapping of ITIL V.3 with 4 Green IT initiatives

4 initiatives contributing to Green ITSM

1. **Procurement** of power efficient hardware is identified by many researchers, for example Blanke (2009). Erek et al. focus on the „source" phase of the IS value chain and the need to audit suppliers for **conformance to energy efficient** standards (2009). Specifically, the use of thin clients in place of PCs is promoted by Velte et al. (2008) and Cole (2003).
2. Servers have proliferated in many data centres and as well as drawing resources for power and cooling, occupy valuable space. **Server consolidation** offers opportunities to consolidate IT resources (Erek et al. 2009; Velte, Velte & Elsenpeter 2008).
3. **Power management** of data centres and PCs is acknowledged as an important initiative (Murugesan 2008).
4. In **view of the toxic chemicals** incorporated into IT equipment, **disposal** is an important challenge. IT managers need to consider decommissioning, recycling or reusing unused or obsolete equipment (Matthews & Matthews 2003; Saied & Velaquez 2003).

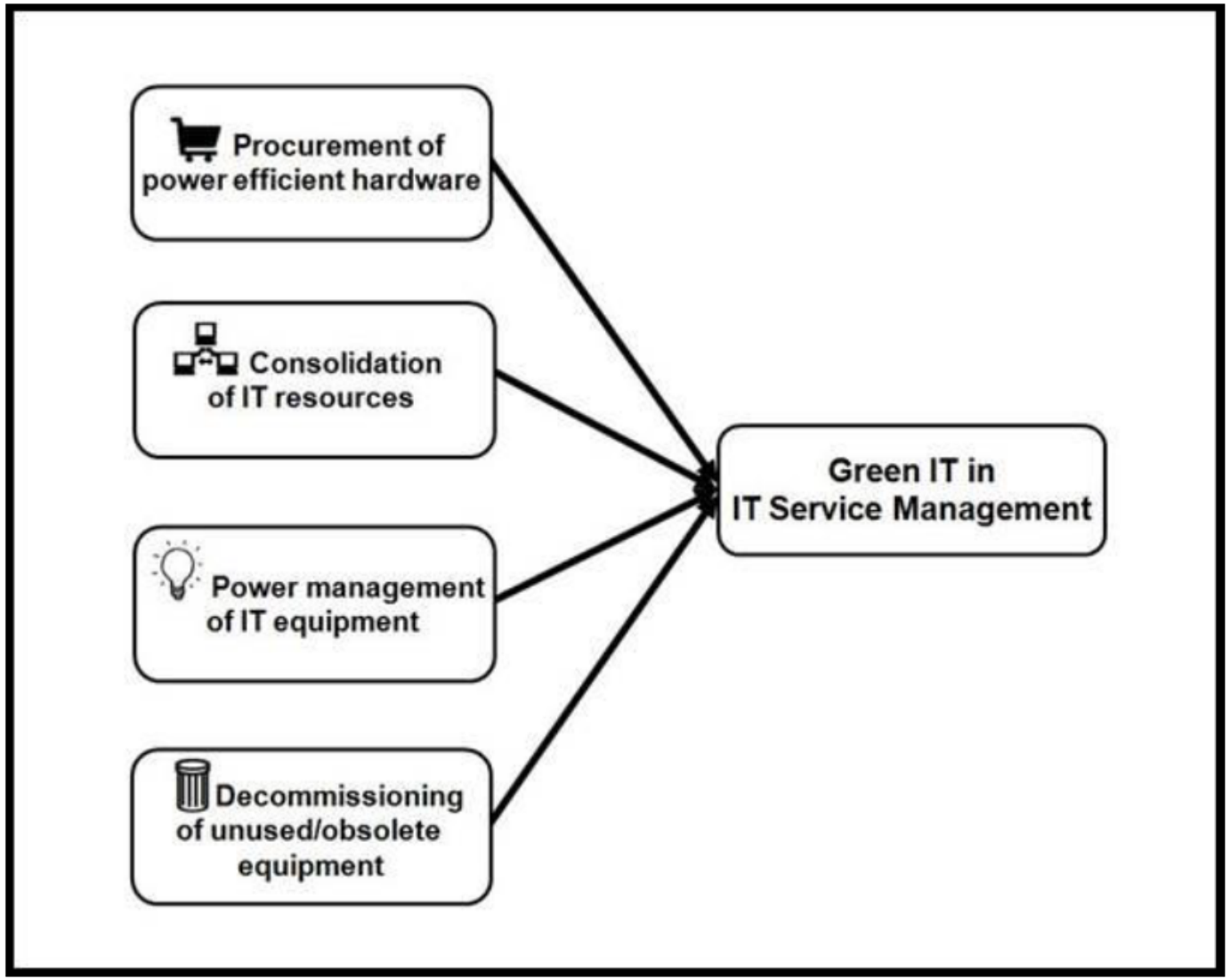

Figure 1. A preliminary framework of initiatives contributing to Green IT Service Management

ITSM Processes/Guidelines				
Financial Management Regulatory and environmental-related planning should get its triggers from within the business. However, Financial Management should apply the proper financial inputs to the related services value, whether cost based or value based. (Section 5.1.2.5)	X	X	X	X
Demand Management Excess capacity generates cost without creating value that provides a basis for cost recovery. Customers are reluctant to pay for idle capacity unless it has value for them. (Section 5.5.1)	X	X	X	

Table 2. Green IT Guidelines in Service Strategy (OGC 2007d)

ITSM Processes/Guidelines				
Environmental Architectures & Standards Every organization should produce an environmental policy for equipment location, with minimum agreed standards for particular concentrations of equipment … Building/site … Major equipment room … Major data centres … Regional data centres and major equipment centres … Server or network equipment rooms … Office environments. (Appendix E: Environmental architectures and standards)			X	
Availability Management A key activity for Availability Management is continually to look at opportunities to optimize the availability of the IT infrastructure in conjunction with Continual Service Improvement activities … The optimization approach is a sensible first step to delivering better value for money. (Section 4.4.5.2)		X	X	

Table 3. Green IT Guidelines in Service Design (OGC 2007b)

Capacity Management The main objective of Component Capacity Management is to identify and understand the performance, capacity and utilization of each of the individual components within the technology used to support the IT services … This ensures the optimum use of the current hardware and software resources in order to achieve and maintain the agreed service level. (Section 4.3.5.3)		X	X	
Service Level Management Service Level Management should include … proactive prevention of service failures, reduction of service risks and improvement in the quality of service, in conjunction with all other processes. (Section 4.2.2)		X	X	X
Supplier Management IT Supplier Management often has to comply with organizational or corporate standards, guidelines and requirements, particularly those of corporate legal, finance and purchasing. (Section 4.7.2)	X			

Table 3. Green IT Guidelines in Service Design (OGC 2007b)

ITSM Processes/Guidelines				
Release & Deployment Management Some specific aspects need to be considered for decommissioning and retiring services and service assets … disposing of equipment according to environmental policies and procedures. (Section 4.4.5.6)				X
Service Asset & Configuration Management Configuration Management … enables other processes to access valuable information, e.g.: optimize asset utilization and costs … consolidate data centres, reduce variations and re-use assets. (Section 4.3.4.2)	X	X	X	X

Table 4. Green IT Guidelines in Service Transition (OGC 2007e)

ITSM Processes/Guidelines				
Server Management & Support The procedures and activities which must be undertaken by the Server Team(s) or department(s) include … procurement advice on the selection, sizing, procurement and usage of servers and related utility software … definition and management of virtual servers … provide information and assistance to Capacity Management to help achieve optimum throughout, utilization and performance from available servers … Decommissioning and disposal of old server equipment. (Section 5.4)	X	X	X	X
Facilities Management … key activity of Facilities Management is to manage the utilization of power … as natural resources become scarcer and expensive, more attention is being focused on techniques to manage utilization more responsibly. (Appendix E: Detailed description of Facilities Management)			X	

Table 5. Green IT Guidelines in Service Operation (OGC 2007c)

ITSM Processes/Guidelines				
The 7-Step Improvement Process CSI will also be interested in determining whether that level of performance can be sustained at a lower cost or whether it needs to be upgraded to an even better level of performance. CSI may therefore also need access to regular performance reports. (Section 4.1)	X	X	X	X

Table 6. Green IT Guidelines in Continual Service Improvement (OGC 2007a)

Recommendations

- CIOs and IT Service managers need to enquire about Green initiatives in their organisation at the Corporate level and engage with the process. **They can use the ITIL 7-step (see last lecture) method to assess the current situation, develop an action plan and then achieve Green credentials for the IT department**. In particular, IT Service managers must consider how to use existing technologies, and work to incorporate new ones into their organisation's Green IT policies and practices. To do this, they need to put pressure on their IT managers and CIOs to move Green benefits to the forefront of their strategic thinking. (Cater Steel)
- As climate change is a global and complex issue, managers need to invest time and funds in training to improve their knowledge. For example, the British Computer Society offers a Green IT certificate which covers Green IT regulations, legislation and policy, identification and base-lining of current Green IT credentials, and advice on how to move forward in delivering Greener IT (BCS 2009).

G-readiness Model

HELP ORGANISATIONS EVALUATE THEIR READINESS FOR ADOPTING GREEN IT. (MOLLA ET AL. 2008)

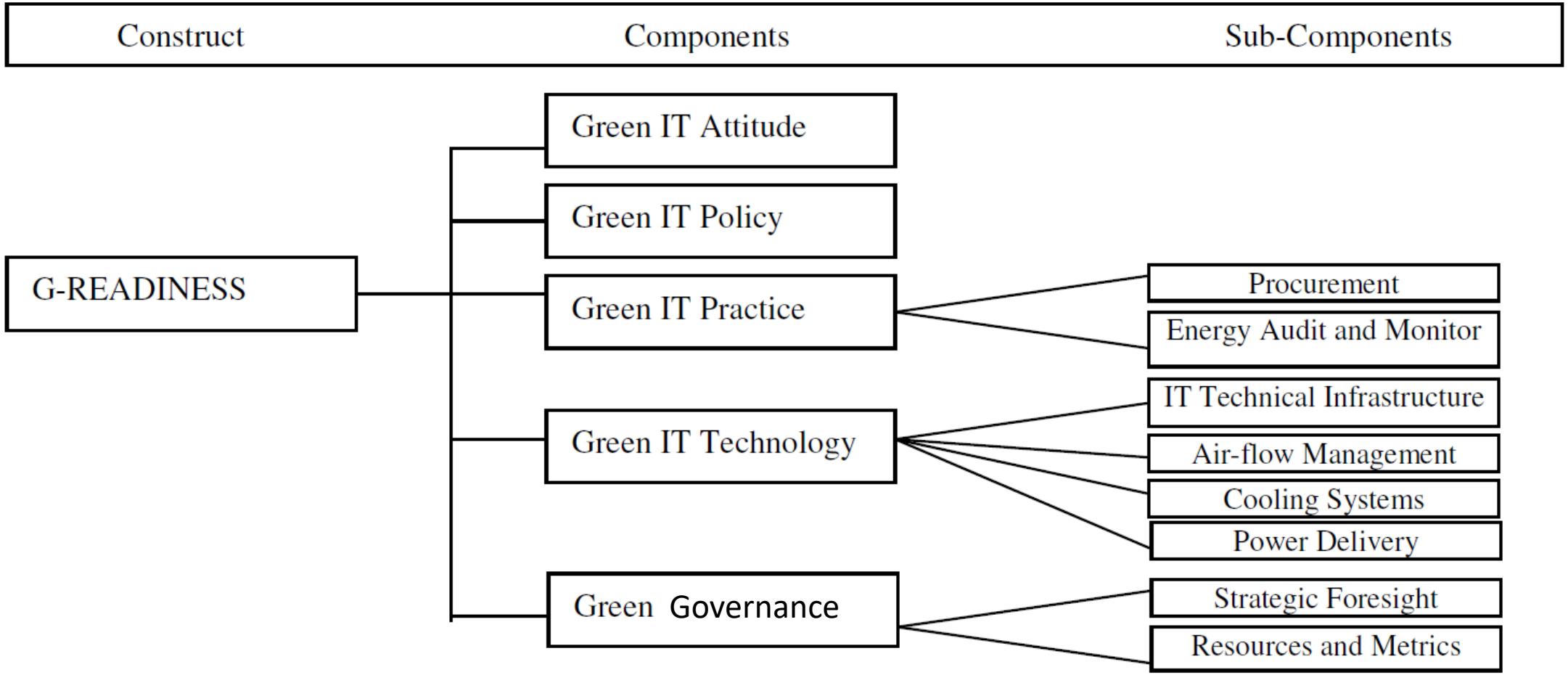

Figure 7. The G-readiness Model

Molla, A, Cooper, V, Corbitt, BJ, Deng, H, Peszynski, KJ, Pittayachawan, S & Teoh, SY 2008, 'E-readiness to G-readiness: developing a green information technology readiness framework', paper presented to ACIS 2008, Christchurch, NZ.

Attitude *Dimension*

- "Organizations are likely to adopt very different attitudes at the corporate level to dealing with sustainability, and these differing attitudes will impact their expectations of Green IT (CFO 2009; Hart 1997; InfoTech 2008a)."
- "*Green IT attitude* is defined as an organization's IT human infrastructure's sentiment towards climate change and eco-sustainability." (Molla et al 2009)
- "It measures the extent to which both IT and business are aware and concerned about the impact of IT on eco-sustainability."

IT Policy *Dimension*

- "Encompasses the frameworks an organization puts in place to apply environmental criteria in its IT - related activities."

- "It measures the extent to which Green issues are encapsulated in organizational procedures guiding the sourcing, use and disposal of the IT technical infrastructure and the activities of the IT human infrastructure." (Goasduff and Forsling 2007; InfoTech 2007a, 2007b, 2008b; Olson 2008).

Practice dimension

- *"Green IT Practice*, pertains to the actual application and realization of sustainability considerations in IT infrastructure sourcing, operation and disposal."

- "Organizations are likely to vary in the actual practice of analysing the Green track record of IT hardware, software, and services providers (CFO 2009; InfoTech 2007c)."

Technological dimension

Refers to technologies and information systems for:

A. Reducing the energy consumption of powering and cooling corporate IT assets (such as data centers)

B. Optimizing the energy efficiency of the IT technical infrastructure

C. Reducing IT induced greenhouse gas emissions

D. Replace carbon emitting business practices

E. Analyzing a business's total environmental footprint

(Accenture 2008; CFO 2009; InfoTech 2007c; 2007d; Chen et al 2008; Elliot and Binny 2008).

Governance dimension

- Is the operating model that defines the administration of Green IT initiatives and is closely related to the policy construct.
- Roles, responsibilities, accountability and control for Green IT initiatives need to be clearly established.
- Businesses should determine whether the responsibility for Green IT initiatives should be assigned to CIO's or to environmental managers

Result (example)

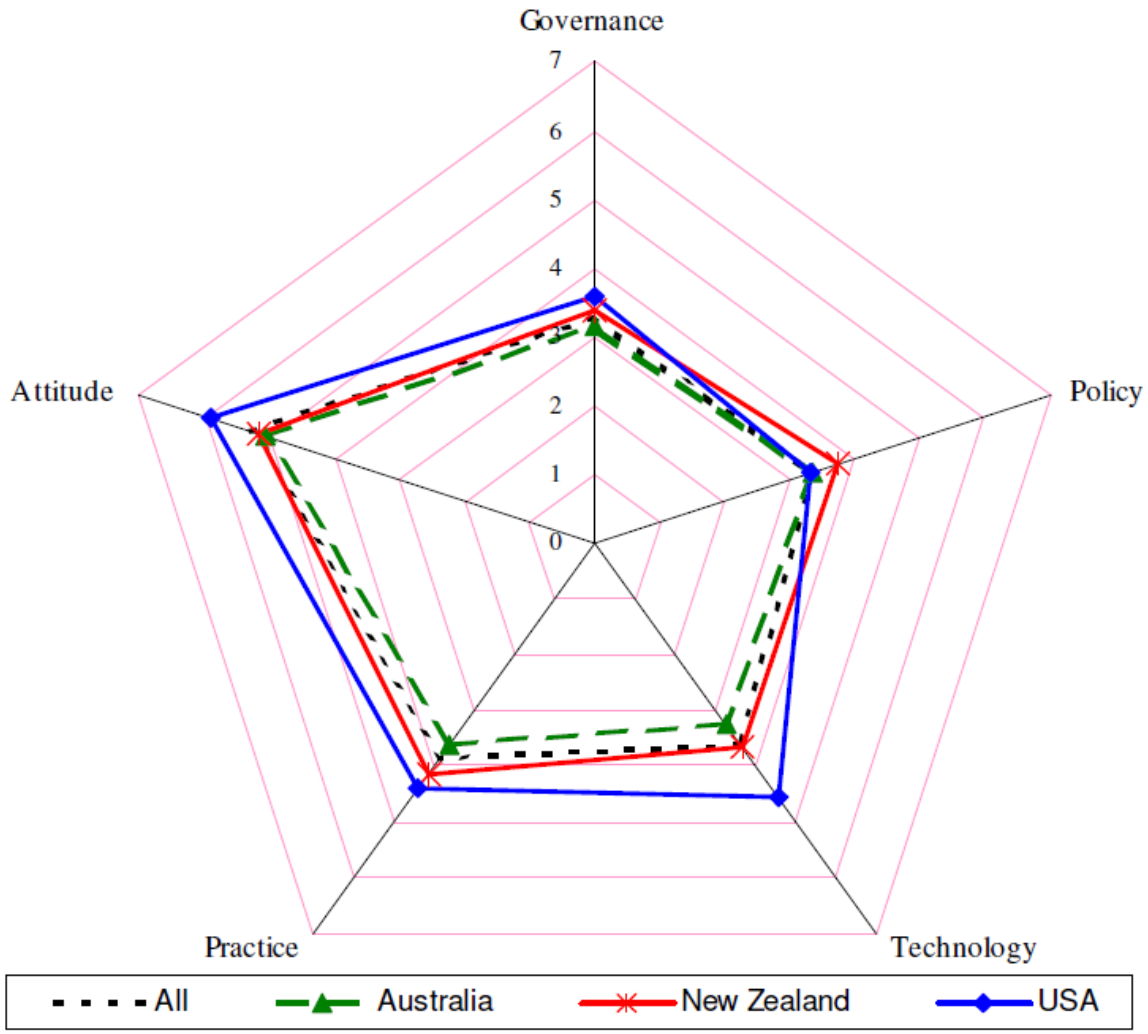

Figure 8. G-readiness Results for Australia, New Zealand and USA